

Logistic Regression for CVD Prediction

ELiud Mule

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Set Working Directory

```
#setwd("~/Users/user/Desktop/Professional DS Mule/R")
```

** Task 1: Data Import and Exploration (10 Marks)**

Load libraries

```
library(tidyverse)

## --- Attaching core tidyverse packages --- tidyverse 2.0.0 ---
## ✓ dplyr      1.2.0   ✓ readr      2.2.0
## ✓ forcats    1.0.1   ✓ stringr  1.6.0
## ✓ ggplot2    4.0.3   ✓ tidbale   3.3.1
## ✓ lubridate  1.9.5   ✓ tidyr     1.3.2
## ✓ purrr      1.2.1
## --- Conflicts --- tidyverse_conflicts() ---
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()   masks stats::lag()
## I Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(dplyr)
library(ggplot2)
library(caret)

## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##   lift

library(psych)

##
## Attaching package: 'psych'
##
## The following objects are masked from 'package:ggplot2':
##
##   %%, alpha

library(caret) #for machine learning models
library(psych) #for description of data
library(ggplot2) #for data visualization
library(caretEnsemble) # enables creation of ensemble models
library(tidyverse) #for data manipulation
library(mlbench) #for benchmarking ML models
library(flextable) #to create and style tables

## Attaching package: 'flextable'
##
## The following object is masked from 'package:purrr':
##
##   compose

library(mltools)

##
## Attaching package: 'mltools'
##
## The following object is masked from 'package:tidyr':
##
##   replace_na

library(tictoc) #for timing code execution
library(ROSE) #for random sampling

## Loaded ROSE 0.0-4

library(ROCR) #for ROC curve
library(smotefamily) # for oversampling
```

Import the dataset

```
data <- read.csv("medical_data.csv")
head(data)

##   age gender      bmi blood_pressure cholesterol glucose smoking alcohol
## 1 36.59287    0 19.88353      126.7126      207.0147 105.81544    1    0
## 2 41.54734    1 32.11152      125.8728      145.6313  76.89136    1    1
## 3 65.38062    1 30.43908      105.3434      156.2852 101.60778    1    0
## 4 46.05763    1 25.46273      140.1740      256.5752 130.69754    0    0
## 5 46.93932    0 26.90125      107.6709      136.6122  58.71636    0    0
## 6 70.72597    0 29.44199      135.1819      285.2456  78.22482    0    0
##   physical_activity family_history stress_level heart_rate sleep_hours
## 1      1.657959          0      3.3274065  77.77227    5.807189
## 2      1.552329          0      4.5888440  79.85721    5.297333
## 3      3.902421          0      0.7929705  77.91827    7.870152
## 4      3.492610          1      1.6643849  83.45057    7.777064
## 5      1.575045          0      2.8040743  80.25018    7.115128
## 6      5.070074          0      1.6687576  74.58525    6.274890
##   medication_adherence urban_residence CVD
## 1                      1              1 No
## 2                      0              1 No
## 3                      1              1 No
## 4                      1              0 No
## 5                      0              0 No
## 6                      1              1 No
```

Explore the dataset

```
str(data)

## 'data.frame':    2500 obs. of  16 variables:
##  $ age      : num  36.6 41.6 65.4 46.1 46.9 ...
##  $ gender    : int   0 1 1 0 0 0 1 1 0 ...
##  $ bmi       : num  19.9 32.1 30.4 25.5 26.9 ...
##  $ blood_pressure : num  127 126 105 140 108 ...
##  $ cholesterol  : num  207 144 156 257 139 ...
##  $ glucose      : num  105.8 76.9 101.6 130.7 58.7 ...
##  $ smoking      : int   1 1 1 0 0 0 0 0 ...
##  $ alcohol      : int   0 1 0 0 0 1 0 1 ...
##  $ physical_activity : num  1.66 1.55 3.9 1.49 1.58 ...
##  $ family_history : int   0 0 1 0 1 0 1 0 ...
##  $ stress_level  : num  3.327 4.559 0.793 1.664 2.804 ...
##  $ heart_rate   : num  77.8 79.9 77.9 83.5 90.3 ...
##  $ sleep_hours  : num  5.81 5.3 7.87 7.78 7.31 ...
##  $ medication_adherence: int   1 0 1 0 1 0 1 1 ...
##  $ urban_residence : int   1 1 1 0 0 1 1 1 ...
##  $ CVD          : chr   "No" "No" "No" "No" "No" ...
```

```
describe(data)

##   vars      n  mean  sd median trimmed  mad   min   max
##  age      1 2500 45.16 14.83 44.94 45.15 14.64 -0.72 95.86
##  gender    2 2500  0.49  0.50  0.00  0.49  0.00  0.00  1.00
##  bmi       3 2500 24.99  5.01 27.08 27.01 4.96 10.55 45.58
##  blood_pressure 4 2500 130.09 15.05 129.92 130.08 15.65 72.32 187.72
##  cholesterol  5 2500 199.80 40.02 199.01 199.72 39.38 63.92 322.70
##  glucose      6 2500 99.65 30.15 99.73 99.87 30.18  2.66 187.58
##  smoking      7 2500  0.29  0.45  0.00  0.24  0.00  0.00  1.00
##  alcohol      8 2500  0.38  0.49  0.00  0.36  0.00  0.00  1.00
##  physical_activity 9 2500  2.93  1.49  2.91  2.92  1.47 -2.13  8.56
##  family_history 10 2500  0.25  0.43  0.00  0.18  0.00  0.00  1.00
##  stress_level 11 2500  4.97  2.01  4.99  4.98  1.97 -1.56 12.84
##  heart_rate   12 2500  74.96 10.15 74.95 74.92 10.63 39.54 118.23
##  sleep_hours  13 2500  6.95  1.49  6.92  6.95  1.51  2.54 12.25
##  medication_adherence 14 2500  0.68  0.47  1.00  0.72  0.00  0.00  1.00
##  urban_residence 15 2500  0.60  0.49  1.00  0.62  0.00  0.00  1.00
##  CVD         16 2500  1.10  0.30  1.00  1.00  0.00  1.00  2.00
##           range skew kurtosis  se
##  age      96.57 0.03 -0.03 0.30
##  gender    1.00 0.03 -2.00 0.01
##  bmi       35.03 -0.03  0.07 0.10
##  blood_pressure 115.40 -0.01  0.04 0.30
##  cholesterol 258.78 0.03 -0.01 0.80
##  glucose    184.92 -0.08 -0.06 0.60
##  smoking     1.00 0.92 -1.15 0.01
##  alcohol     1.00 0.48 -1.77 0.01
##  physical_activity 10.68 0.10  0.03 0.03
##  family_history  1.00 1.17 -0.63 0.01
##  stress_level 14.40 -0.01 -0.01 0.04
##  heart_rate   78.69 0.03 -0.06 0.20
##  sleep_hours  9.71 0.04 -0.13 0.03
##  medication_adherence 1.00 -0.77 -1.41 0.01
##  urban_residence 1.00 -0.39 -1.85 0.01
##  CVD         1.00 2.64  4.99 0.01
```

Check for missing values per variable

```
data.frame(sapply(data, function(x) sum(is.na(x))))
```

```
##           sapply.data..function.x.sum.is.na.x...
##  age              0
##  gender            0
##  bmi               0
##  blood_pressure    0
##  cholesterol       0
##  glucose           0
##  smoking           0
##  alcohol           0
##  physical_activity 0
##  family_history    0
##  stress_level      0
##  heart_rate        0
##  sleep_hours       0
##  medication_adherence 0
##  urban_residence  0
##  CVD               0

library(sjmisc)

##
## Attaching package: 'sjmisc'

##
## The following object is masked from 'package:mltools':
##
##   replace_na

##
## The following object is masked from 'package:purrr':
##
##   is_empty

##
## The following object is masked from 'package:tidyr':
##
##   replace_na

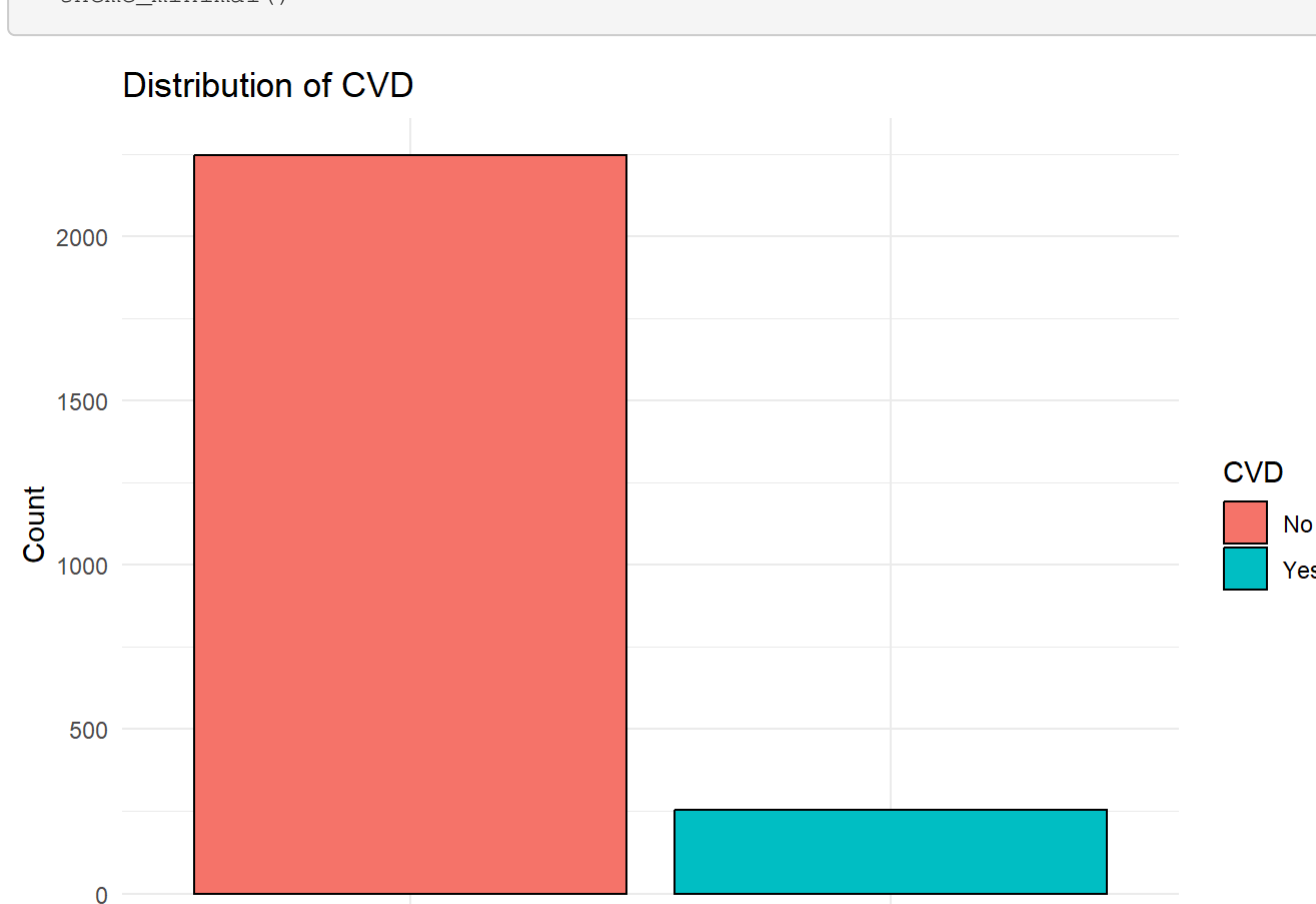
##
## The following object is masked from 'package:tidbale':
##
##   add_case
```

```
freq(data, CVD)
```

```
## CVD <character>
## # total N=2500 valid N=2500 mean=1.10 sd=0.30
##
## Value |    N | Raw % | Valid % | Cum. %
## -----
## No    | 2247 | 89.88 | 89.88 | 89.88
## Yes   | 253  | 10.12 | 10.12 | 100.00
## <NA>  |  0   | 0.00 | <NA> | <NA>
```

Visualize the distribution of the target variable CVD

```
ggplot(data, aes(x = CVD, fill = CVD)) +
  geom_bar(color = "black") +
  labs(title = "Distribution of CVD", x = "CVD Status", y = "Count") +
  theme_minimal()
```



Task 2: Data Preprocessing (15 Marks)

Convert categorical variables to factors

```
data$gender <- as.factor(data$gender)
data$smoking <- as.factor(data$smoking)
data$alcohol <- as.factor(data$alcohol)
data$urban_residence <- as.factor(data$urban_residence)
data$family_history <- as.factor(data$family_history)
data$medication_adherence <- as.factor(data$medication_adherence)
```

** Task 3: Data Partitioning (10 Marks)**

Split the dataset into training and testing sets

```
set.seed(2026)
train_index <- createDataPartition(data$CVD, p = 0.7, list = FALSE)
train_data <- data[train_index, ]
test_data <- data[-train_index, ]
```

Class distribution in both training sets

```
dim(train_data)
```

```
## [1] 1751 16
```

```
dim(test_data)
```

```
## [1] 749 16
```

** Task 4: Model Training with trainControl (20 Marks)**

Set up trainControl for 10-fold cross-validation

```
train_control <- trainControl(method = "cv", number = 10, classProbs = TRUE, summaryFunction = twoClassSummary)
```

Train the logistic regression model

```
library(tictoc)
tic()
set.seed(123) # Set seed for reproducibility
model <- train(CVD ~ ., data = train_data,
              method = "glm",
              preProcess = c("scale", "center"),
              trControl = train_control)

## Warning in train.default(x, y, weights = w, ...): The metric "Accuracy" was not
## in the result set. ROC will be used instead.

toc()

## 1.11 sec elapsed
```

View Model

```
summary(model)

##
## Call:
## glm2
##
## Coefficients:
## (Intercept)      0.41255    0.16002 -20.680 < 2e-16 ***
## age           0.56981    0.09780  5.826 5.66e-09 ***
## gender1      0.19928    0.09600  2.076 0.037918 *
## bmi          0.45684    0.09701  4.709 2.48e-06 ***
## blood_pressure 0.03364    0.09693  3.442 0.000577 ***
## cholesterol  0.61037    0.10228  5.968 2.41e-09 ***
## glucose      0.76367    0.10290  7.421 1.16e-13 ***
## smoking1     0.37949    0.09089  4.175 2.98e-05 ***
## alcohol1     0.11137    0.09398  1.185 0.236025
## physical_activity -0.80608    0.10749 -7.499 6.42e-14 ***
## family_history1 0.47410    0.08739  5.425 5.79e-08 ***
## stress_level  0.67056    0.10063  6.664 2.67e-11 ***
## heart_rate    -0.02897    0.09362 -0.309 0.756965
## sleep_hours   -0.83066    0.10709 -7.826 5.06e-15 ***
## medication_adherence -0.35754    0.09151 -3.907 9.33e-05 ***
## urban_residence -0.03740    0.09440 -0.396 0.691943
## ---
## Signif. codes:  0 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1151.13 on 1750 degrees of freedom
## Residual deviance: 767.96 on 1735 degrees of freedom
## AIC: 799.96
##
## Number of Fisher Scoring iterations: 7
```

View the results above

```
library(sjPlot)

##
## Attaching package: 'sjPlot'

##
## The following object is masked from 'package:ggplot2':
##
##   set_theme

tab_model(model$finalModel,
          show.intercept = TRUE,
          # show.ci = FALSE,
          show.se = TRUE,
          show.stat = TRUE,
          show.r2 = TRUE)

##
## Dependent variable
## Predictors Odds Ratios Std. Error CI Statistic p
## (Intercept) 0.03 0.01 0.02 - 0.04 -20.68 <0.001
## age 1.77 0.17 1.46 - 2.15 5.83 <0.001
## gender1 1.22 0.12 1.01 - 1.48 2.08 0.038
## bmi 1.58 0.15 1.31 - 1.91 4.71 <0.001
## blood pressure 1.40 0.14 1.16 - 1.69 3.44 0.001
## cholesterol 1.84 0.19 1.51 - 2.26 5.97 <0.001
## glucose 2.15 0.22 1.76 - 2.64 7.42 <0.001
## smoking1 1.46 0.13 1.22 - 1.75 4.18 <0.001
## alcohol1 1.12 0.11 0.93 - 1.34 1.18 0.236
## physical activity 0.45 0.05 0.36 - 0.55 -7.50 <0.001
## family history1 1.61 0.14 1.35 - 1.91 5.43 <0.001
## stress level 1.96 0.20 1.61 - 2.39 6.66 <0.001
## heart rate 0.97 0.09 0.81 - 1.17 -0.31 0.757
## sleep hours 0.43 0.05 0.35 - 0.53 -7.83 <0.001
## medication adherence1 0.70 0.06 0.58 - 0.84 -3.91 <0.001
## urban residence1 0.96 0.09 0.80 - 1.16 -0.40 0.692
## Observations 1751
## R2 Tjur 0.299
```

** Task 5: Model Evaluation (25 Marks)**

Make predictions on the test set

```
pred_class <- predict(model, test_data)
pred_prob <- predict(model, test_data, type = "prob")
```

```
test_data$CVD <- factor(test_data$CVD, levels = c("No", "Yes"))
pred_class <- factor(pred_class, levels = c("No", "Yes"))
```

Confusion Matrix

```
conf_matrix <- confusionMatrix(pred_class, test_data$CVD, positive = "Yes")
print(conf_matrix)

## Confusion Matrix and Statistics
##
## Reference
## Prediction No Yes
## No 661 60
## Yes 13 15
##
## Accuracy : 0.9025
## 95% CI : ( 0.879, 0.9228)
## No Information Rate : 0.9999
## P-Value (Acc > NIR) : 0.4338
##
## Kappa : 0.2505
##
## Mcnemar's Test P-Value : 7.289e-08
##
## Sensitivity : 0.20000
## Specificity : 0.98071
## Pos Pred Value : 0.53571
## Neg Pred Value : 0.91678
## Prevalence : 0.10013
## Detection Rate : 0.02003
## Detection Prevalence : 0.03738
## Balanced Accuracy : 0.59036
##
## 'Positive' Class : Yes
```

plot ROC curve and find area under the curve

```
library(pROC)

## Type 'citation("pROC")' for a citation.

##
## Attaching package: 'pROC'

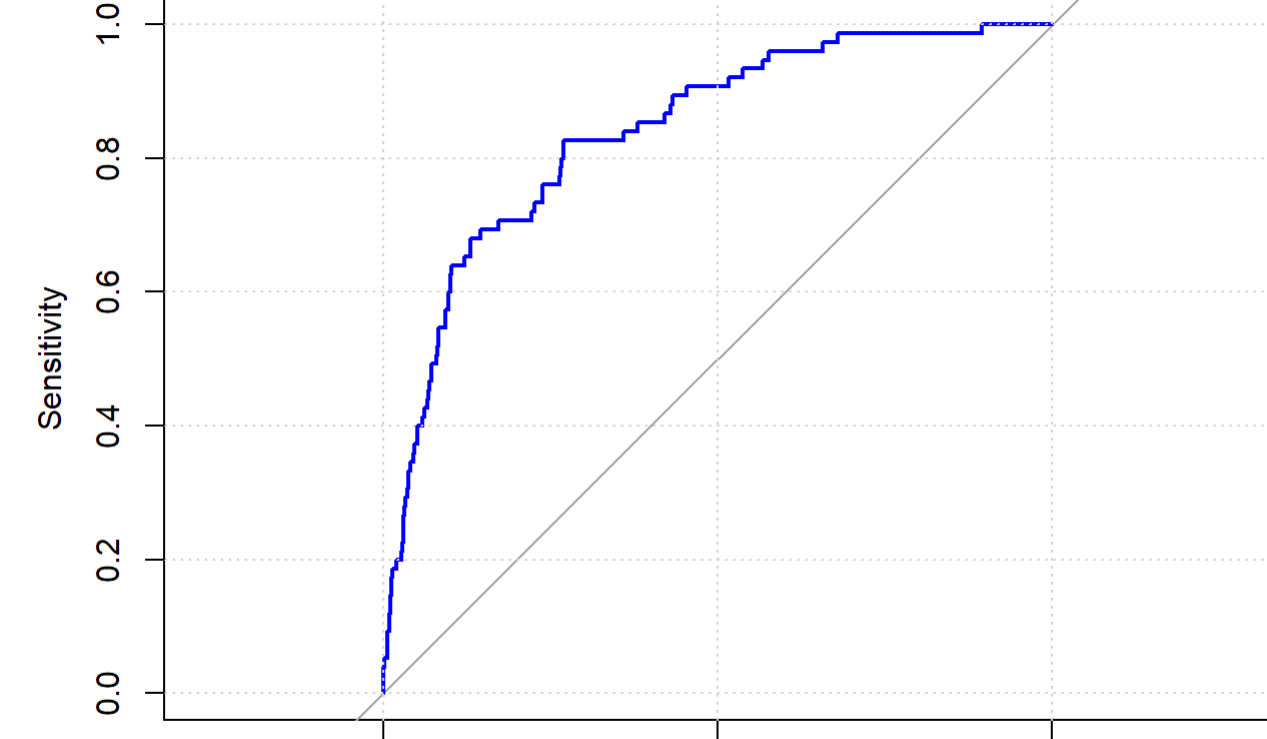
##
## The following objects are masked from 'package:stats':
##
##   cov, smooth, var

roc_obj <- roc(test_data$CVD, pred_prob[, "Yes"])

## Setting levels: control = No, case = Yes

## Setting direction: controls < cases

plot(roc_obj, col = "blue", main = "ROC Curve for CVD Prediction", lwd = 2)
grid()
```



```
auc_value <- auc(roc_obj)
print(paste("AUC:", auc_value))
```

```
## [1] "AUC: 0.8395450049455598"
```